

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Design of a Novel Miniature MALDI-TOF Mass Spectrometer for High Throughput Medical Screening

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ABSTRACT

Our laboratory has been developing a series of mid-resolution miniature MALDI-TOF mass spectrometers for the detection of biomarker signatures. Applications benefiting from these devices include high throughput biological screening assays and real-time biological weapons detection. As we continue to decrease the overall size of our designs, several fundamental parameters become increasingly important to achieving suitable resolving power. Examination of these fundamental parameters has led us to a novel design having potentially greater performance than is possible with conventional linear TOF instruments. In this presentation we will discuss the fundamental improvements required for enhanced performance and our current progress in implementing this new design.